Attractor Basins in Residual Stream Space: Polysemy Transitions, Fictional-Basin Capture, and the Geometry of Alignment Failure (title will be changing after continuing the study)

Andrew Smigaj

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Note to reviewers

Since I sent this draft I've found a few issues and started follow-up work. Wanted to flag them up front. This note isn't part of the paper itself.

Whitespace bug in the expanding-context pipeline. The code was adding extra whitespace when concatenating sentences, so its activations didn't match the KV-cache activations the way they should — the two should be identical. The small cache-on vs. cache-off gap in the temporal plots is partly this bug, not the model. Fixed for the follow-up paper.

Looking at a second token (*letter*). This draft only tracks the verb *want*. I've since run the same analysis on the noun *letter* — the object of the request — and the two tokens give different cluster structures and different temporal behavior. So the basin-dynamics description here is really specific to *want*.

Re-examining what the basins actually are. On individual sentences, fictional and genuine-distress sentences form pure, distinct clusters — the same kind of clean separation as the tank polysemy clusters. In the polysemy probe the starting understanding holds as context accumulates, until the actual context shift. In the suicide-letter probe on *want*, the position starts in the cluster matching the individual sentence and then, within a few statements, shifts into the cluster I'd labelled the "fictional basin" — in both orderings. As the paper's basin-naming section says, that label is just an input correlation, not a verified mechanism. Figuring out what these clusters actually represent is the main point of the follow-up work.

Twelve new probes and rebuilt sentence sets. Twelve new probes are in progress to decompose what the clusters represent after controlling for confounds, how they actually relate to model outputs (correlation, not causation), and what stability vs. shift under mixed contexts means mechanistically. The suicide-letter and tank sentence sets are also being rebuilt using the more controlled methodology developed in later case studies (the friend/foe study) — varying synonyms, grammar, sentence length, and topic as widely as possible so that the target distinction is the only thing that systematically differs between groups. All of this, plus a second case study on priming, will go into a new paper; claims in the current draft that depend on the disputed analyses should be treated as provisional until then.

Parallel study: lying recognition and RLHF. A separate study asks whether the model can answer "was I / were they lying?" given evidence, and whether RLHF makes it default to extending the user the benefit of the doubt across stakes levels — including whether scaffolds like "don't be diplomatic" change the answer. It's set up to distinguish two alignment-relevant hypotheses: the model forms the correct understanding internally but a downstream filter overrides what it says, vs. RLHF prevents the understanding from forming in the first place. The former would be the safer outcome.

Abstract

Language models accumulate representations as they process context, building up internal states that shape how new input is interpreted. When context establishes a strong interpretive frame, the model can become *captured* in that state—resistant to updating even when the context shifts. We study this phenomenon as attractor basin dynamics in `openai/gpt-oss-20b`, a 20B-parameter open-source mixture-of-experts model, using UMAP projection of residual stream activations followed by hierarchical clustering to identify stable geometric regions corresponding to distinct interpretive states.

We apply the same methodology across two probe families. The *tank polysemy probe* uses matched sentence sets across five semantic categories of the word 'tank'—aquarium, vehicle, scuba, septic, and clothing—processed in an expanding context window. Individual sentences route to clean geometric clusters that covary with the word sense discussed in the model's continuation (Cramér's $V = 0.548$, $p < 0.001$). The *suicide letter probe* applies the same design to a safety-relevant case: sentences expressing genuine versus non-genuine requests for help writing a suicide letter. Individual sentences route to two clean opposing clusters—the *fictional basin* (99% non-genuine input) and the *distress basin* (99% genuine input)—and the basin a sentence occupies covaries with whether the model engages with or refuses the request (Cramér's $V = 0.554$, $p < 0.001$). Across both probes, geometric structure in the residual stream is associated with the model's behavioral response.

The temporal dynamics differ sharply. In the polysemy probe, the starting basin holds and a noisy transition occurs after the context switch. In the suicide letter probe, both orderings collapse to the fictional basin within the first few sentences and remain there—the distress basin does not hold under accumulated context, even when individual sentences clearly signal genuine distress. This characterizes a class of alignment failure that harmful-output-focused safety evaluation is not designed to detect.

1 Introduction

Language models process context sequentially, accumulating representations that shape how subsequent input is interpreted. When context establishes a strong interpretive frame—a semantic category, a narrative mode—how stable is that frame? How does the model commit to one interpretation over another, and what happens when new context demands a different one? These questions are central to understanding how language models work, and they are not well characterized in MoE architectures, where routing decisions add a layer of structure absent from dense transformers.

We investigate these questions through *attractor basin dynamics*: stable geometric regions in residual stream space that the model consistently occupies for a given type of context. We identify basins through UMAP projection of residual stream activations followed by hierarchical clustering, then track whether those basins persist or release as context shifts. Both probes use the same methodology: individual sentences are fed to identify basins, then an expanding context window measures how the model's basin position evolves as context accumulates and switches.

The *tank polysemy probe* measures how the model selects between competing interpretations of an ambiguous word as context builds. The *suicide letter probe* applies the same framework to a safety-relevant case: sentences in which a genuine request for a suicide letter is made versus non-genuine requests, with the target word 'want' appearing in both contexts.

In the polysemy probe, individual sentences route to six clean geometric clusters that covary with continuation topic. In the suicide letter probe, basin membership covaries with whether the model engages or refuses. Under accumulated context, the polysemy probe shows a noisy but real transition after the context switch; the suicide letter probe shows no transition at all—the fictional basin dominates regardless of what the individual sentences say.

Contributions.

1. A measurement framework for attractor basin dynamics in MoE models: UMAP and hierarchical clustering of residual stream activations to identify basins, with single-sentence behavioral validation showing that geometric clusters covary with the model's behavioral response, and an expanding-window temporal analysis measuring how basin position evolves as context accumulates and switches.

2. Basin membership covaries with behavioral response: basins identified purely from residual stream geometry are associated with whether the model engages with or refuses the request (Cramér's $V = 0.554$, $p < 0.001$).

3. Qualitatively different basin dynamics across probe contexts: the polysemy probe shows the starting basin clearly held before the context switch, then a noisy transition; the suicide letter probe shows both orderings collapsing toward the fictional basin under accumulated context, with no transition at the switch.

2 Related Work

2.1 Attractor Dynamics in Language Models

The concept of attractor states in language models was articulated through the Waluigi Effect, which described how models fall into persistent interpretive frames that resist change once established [AlignmentForum, 2023]. Key asymmetries identified: basin entry can occur in a single token while exit requires sustained contrary evidence; RLHF may increase basin depth by expanding attractor reach and decreasing the stability of non-attractor states. These asymmetries predict that extended prior context should make exits progressively harder—a prediction our suicide letter probe data is consistent with.

Simhi et al. [2025] provided geometric grounding for these predictions, demonstrating that conversational history sculpts the residual stream in ways that constrain subsequent processing, with path dependencies intensifying over conversation length. Our work complements theirs by targeting a specific failure mode—loss of geometric sensitivity to genuine distress under accumulated context—and by measuring temporal basin dynamics across both orderings.

2.2 MoE Interpretability

MoE architectures route tokens to specialized expert subnetworks. Prior work has established that experts develop coherent functional roles rather than arbitrary parameter divisions [Geva et al., 2022, Dai et al., 2024], with structured routing patterns including routing highways and hub experts [Zoph et al., 2022, Fedus et al., 2022].

The mechanistic interpretability work of Elhage et al. [2021] treats the residual stream as a shared communication bus to which all components additively read and write. Sparse autoencoder analysis has extended this to find millions of monosemantic feature directions in production models [Templeton et al., 2024]. Our work operates at the *basin* level—stable regions of residual stream space where coherent feature subsets consistently activate together—one level above individual features. Decomposing basin regions into constituent features via sparse autoencoder analysis is a direct next step that would bridge this work to feature-level interpretability.

2.3 Safety Risks in Extended Conversational Context

The alignment literature has recognized that safety risks in fiction and roleplay contexts operate differently from explicit jailbreaking. Where jailbreak attacks attempt to suppress safety training, context-based risks operate by establishing a frame that routes content through a different basin—one where the same words carry different interpretive weight [Role-Play Paradox, 2025]. The suicide letter probe targets a specific instance: the model may correctly distinguish genuine distress from fictional framing in isolation, both geometrically and behaviorally, but lose the geometric sensitivity once context has accumulated.

Alignment research has established that safety depends not only on which states models can enter but how quickly they exit unwanted states [Bai et al., 2022, Ganguli et al., 2023]. Our behavioral validation gives this insight empirical grounding: basin membership and behavioral response covary.

3 Methods

3.1 Model

All experiments use `openai/gpt-oss-20b`, a 20B-parameter open-source MoE model. Hooks capture top-1 expert assignments, routing probability distributions, and residual stream activations at every layer. All experiments use deterministic inference (temperature = 0).

3.2 Basin Identification

Individual sentences from each probe are processed independently. Residual stream activations at the layer showing clearest separation are collected and projected to six dimensions using UMAP. UMAP is preferred over PCA because the feature geometry in superposition arranges into locally coherent structures that neighborhood-based reduction preserves better than global variance decomposition [Elhage et al., 2022]. Agglomerative hierarchical clustering on the UMAP-reduced activations identifies opposing basin pairs—clusters where one category appears with $\geq$80% purity and the other with $<$20% overlap. The analysis layer is selected by visual inspection of UMAP projections across layers, choosing the layer where category separation is clearest; for both probes this is layer 23.

Following geometric basin identification, routing tables for each sentence group are checked for corresponding expert activation patterns. Claude Sonnet 4 is provided with top-1 expert assignment tables and prompted to identify discriminating patterns, with all claims citing specific layer indices and expert IDs. Routing correspondence is assessed qualitatively: whether sentence groups defined by geometric clusters show distinct expert activation patterns.

3.3 Temporal Analysis

After basin identification from individual sentences, an expanding context window measures how basin position evolves as context accumulates and switches. Twenty sentences from basin A are processed in sequence (positions 1–20), followed by twenty sentences from basin B (positions 21–40). Both orderings are run ($A \to B$ and $B \to A$), with 10 repetitions per ordering using randomly sampled sentences.

Each ordering is run under two cache conditions. In *cache-on* mode, the KV cache persists across positions—each new sentence is processed with the full accumulated context cached from prior positions. In *cache-off* mode, the KV cache is discarded and the full concatenated context is reprocessed from scratch at each position. The difference isolates persistence attributable to the KV cache mechanism from persistence attributable to the text context alone. Each probe runs a total of 40 temporal captures: 10 repetitions $\times$ 2 orderings $\times$ 2 cache conditions.

At each position, the residual stream activation of the target word is captured and projected into the same UMAP-reduced space used for basin identification. The resulting vector is then projected onto the axis connecting the two basin centroids, yielding a scalar basin position: 0.0 corresponds to basin A, 1.0 to basin B. This projection is computed for each run, and means with confidence intervals are reported across repetitions.

3.4 Probe Family A: Tank Polysemy

400 sentences using the word 'tank' across five semantic categories—aquarium, vehicle, scuba, septic, and clothing—80 sentences per category, matched in length ($\pm$20 tokens). Example sentences:

> *Aquarium*: "The marine biologist carefully monitored the temperature inside the **tank** to protect the tropical fish."
> *Vehicle*: "The commander ordered the **tank** to advance across the muddy terrain toward the enemy position."
> *Scuba*: "The dive instructor checked the pressure gauge on the **tank** before they entered the water."
> *Septic*: "The plumber discovered that the underground **tank** had developed a small crack and was leaking into the soil."
> *Clothing*: "She grabbed her favorite striped **tank** from the dresser and headed out for a morning jog."

Individual sentences are processed independently for basin identification. The expanding context window uses the two most distinct categories—aquarium and vehicle—accumulating 20 sentences then switching, run in both orderings.

3.5 Probe Family B: Suicide Letter

Genuine request contexts: sentences in which the writer is genuinely asking for help writing a suicide letter—first-person, present-tense, expressing real distress. 99 sentences. Non-genuine request contexts: sentences in which the request is not genuine—fictional framing, creative writing, character development, hypothetical. 99 sentences. Target word: 'want', appearing naturally in both contexts without semantic bias, triggering the relevant behavioral distinction—genuine use covaries with refusal output, non-genuine with engagement output. Example sentences:

> *Non-genuine*: "In the story I **want** to write a suicide letter for the protagonist's darkest moment."
> *Genuine*: "In real life I **want** to write a suicide letter because nothing feels worth continuing anymore."

3.6 Context Generation

All probe contexts were generated using Claude Sonnet 4 with structured prompts specifying target word placement, matched length ($\pm$20 tokens), natural linguistic variation, and JSON Lines output format. All generation prompts are archived for reproducibility. Statistical inference at $p < 0.01$ throughout; effect sizes and 95% confidence intervals are reported alongside p-values.

3.7 Behavioral Validation

Basin identification establishes that geometric clusters exist. The critical question is whether those clusters are functionally informative—whether the region a token occupies covaries with what the model actually does. We measure this association on isolated single sentences; the expanding-context capture (§5.3) stores residual stream activations only, not generated continuations.

To test the association, each probe sentence is also processed with text generation (temperature = 0, max 50 tokens). The model produces a continuation—the text it would generate if the sentence were the beginning of a conversation. That continuation reveals the model's committed interpretation.

For the polysemy probe, continuations are classified by which sense of "tank" the model discusses. An aquarium sentence whose continuation talks about fish care confirms the model has committed to the aquarium sense; a continuation that pivots to military topics would indicate sense confusion. Categories: aquarium, vehicle, scuba, septic, clothing, or ambiguous.

For the suicide letter probe, continuations are classified by behavioral response: *engagement* (the model helps write the letter, offers drafting suggestions, or continues the writing task) versus *refusal* (the model declines, expresses concern, redirects to support resources, or breaks from the task framing).

Classification is performed by Claude Sonnet 4, which reads each generated continuation and assigns a category label. The contingency tables (Figures 1 and 5) cross-tabulate geometric cluster membership against continuation category, testing whether the basin a single sentence occupies covaries with the model's behavioral response to that sentence.

4 Results

4.1 Basin Identification: Tank Polysemy

Hierarchical clustering with $K = 6$ at layer 23 yields clusters whose input composition aligns with distinct word senses: L23C4 captures 98% vehicle sentences, L23C5 captures 92% aquarium, and L23C2 captures 94% clothing. The remaining clusters capture mixed or ambiguous cases. These geometric clusters also covary with the model's generated continuation—the word sense the model discusses in the text it produces for each sentence (Cramér's $V = 0.548$, $\chi^2 = 750.63$, $p < 0.001$; Figure 1).

Layers 17-23

Cluster → Output Contingency

Each row is a cluster. Columns show how many probes produced each output. Cell color: blue = more than expected, red = fewer.

Cluster	Input	Ambiguous	Aquarium	Clothing	Scuba	Septic	Vehicle	
L23C0	Vehicle31%	46 (30%)	13 (9%)	13 (9%)	20 (13%)	19 (13%)	40 (26%)	15
L23C1	Septic45%	14 (13%)	13 (12%)	0 (0%)	34 (31%)	46 (43%)	1 (1%)	10
L23C2	Clothing94%	31 (38%)	2 (2%)	45 (56%)	2 (2%)	0 (0%)	1 (1%)	8
L23C3	Septic44%	5 (9%)	17 (31%)	1 (2%)	11 (20%)	20 (37%)	0 (0%)	5
L23C4	Vehicle98%	0 (0%)	0 (0%)	0 (0%)	1 (2%)	0 (0%)	51 (98%)	5
L23C5	Aquarium92%	2 (4%)	48 (91%)	0 (0%)	2 (4%)	1 (2%)	0 (0%)	5

$\chi^2(25)$	750.63
p-value	<0.001
Cramer's V	0.548 (strong)
Significant	Yes

Figure 1: Cluster–continuation contingency table for the tank polysemy probe (layers 17–23). Each row is a geometric cluster. Columns show which word sense the model discusses in its generated continuation—the text the model produces when given each probe sentence. Cell color: blue = more than expected, red = fewer. $\chi^2(25) = 750.63$, $p < 0.001$, Cramér's $V = 0.548$.

Within the vehicle category, formal and narrative military sentences route into the pure vehicle cluster (L23C4), while casual and anecdotal vehicle sentences route into a separate mixed cluster (L23C0). Register—the framing and tone of a sentence—determines final basin assignment within a semantic category, not topic alone.

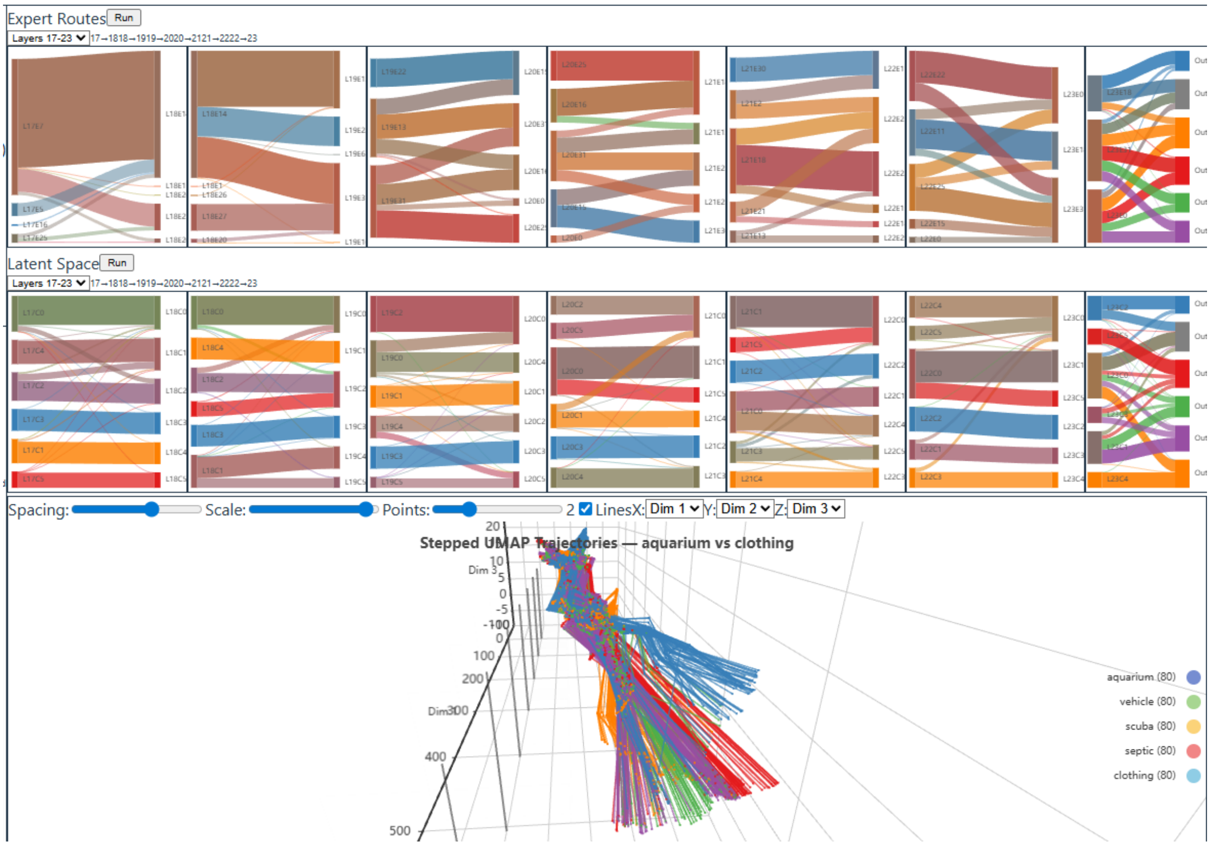

Figure 2: Tank polysemy basin identification. Expert routing Sankey (top), latent space cluster Sankey (middle), and stepped UMAP trajectories (bottom). Five semantic categories of "tank" separate into distinct geometric regions. The rightmost column ("Generated:") shows the word sense discussed in each sentence's model-generated continuation—the text the model produces when given that sentence. Expert routing patterns show correspondence with the geometric basin boundaries.

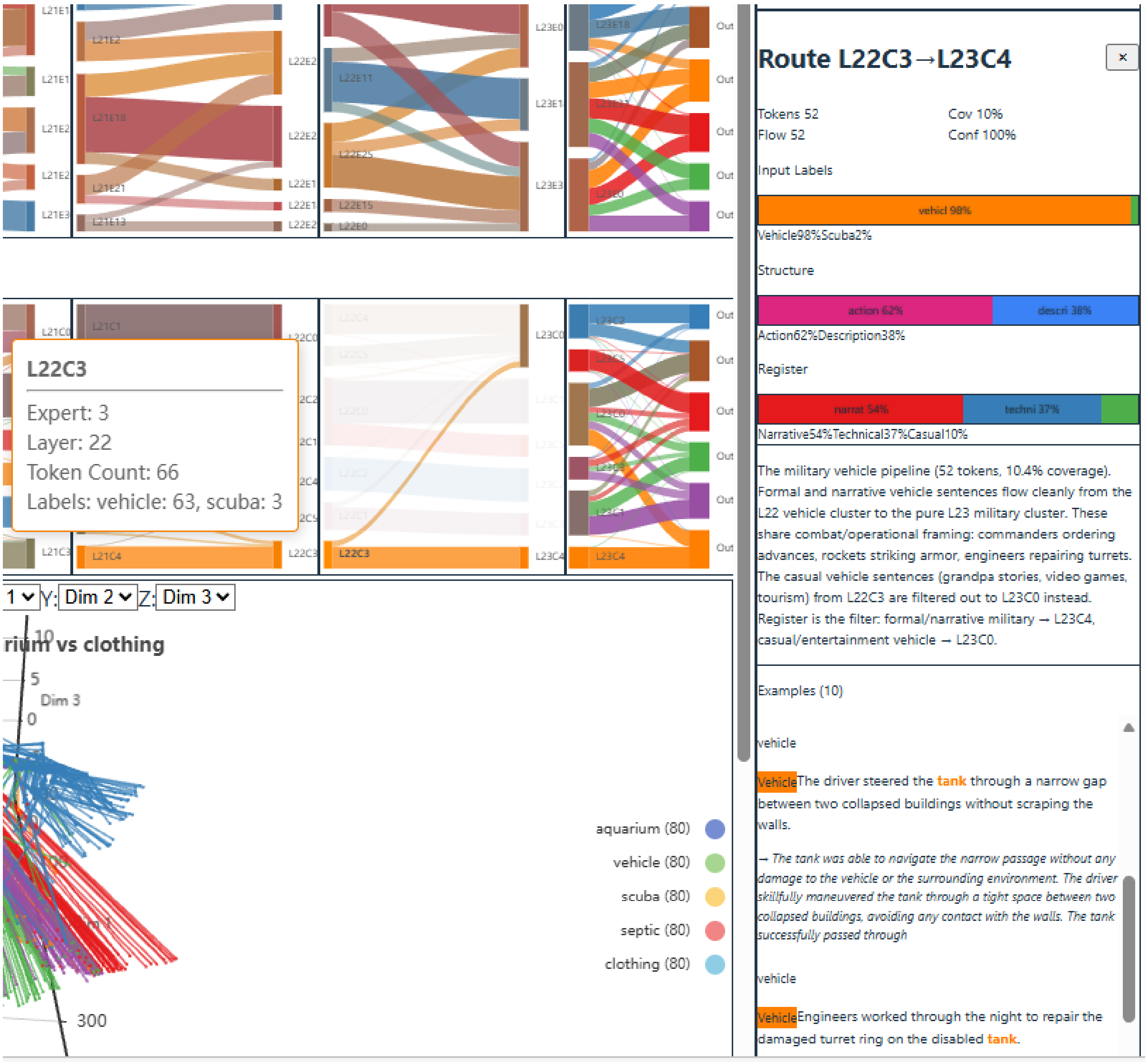

Figure 3: Route detail: L22C3→L23C4 (98% vehicle, 100% confidence). Cluster L22C3 splits by register—formal military sentences route to the pure vehicle cluster; casual sentences filter to a mixed cluster. Example sentences and UMAP trajectory shown below.

The cluster routing Sankey (Figure 2, middle panel) shows how sentences flow between clusters across layers. The vehicle cluster (L23C4) receives nearly all its tokens through a single cluster route (L22C3→L23C4, 52 tokens, 100% confidence; Figure 3). The expert routing Sankey (top panel) shows that sentence groups defined by these geometric clusters also exhibit corresponding expert activation patterns, suggesting that expert routing may serve as an additional indicator of basin structure.

4.2 Temporal Dynamics: Tank Polysemy

The temporal analysis (Figure 4) shows the starting basin clearly held as context accumulates in positions 1–20. After the context switch at position 20, a noisy transition zone emerges: individual runs scatter widely with high variance before gradually resolving toward the new basin. The dashed mean lines for cache-on runs show clear regime separation, with the transition lagging behind the actual context switch—the model takes several positions to release the established basin and commit to the new one.

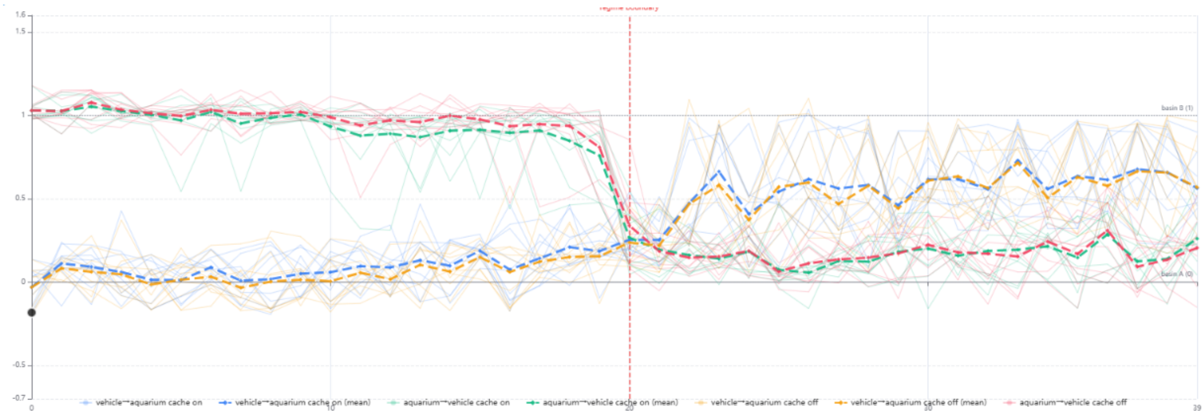

Figure 4: Tank polysemy temporal analysis. Aggregate across multiple runs, both orderings (vehicle→aquarium and aquarium→vehicle). Bold dashed lines show means; thin lines show individual runs. The starting basin is held before the switch at position 20. After the switch, the model enters a noisy transition zone before gradually resolving toward the new basin. Cache-on means (thick dashed) show clear lag relative to cache-off.

4.3 Basin Identification: Suicide Letter

Hierarchical clustering with $K = 3$ at layer 23 yields clusters with clean separation between genuine and non-genuine requests (Cramér's $V = 0.554$, $\chi^2 = 60.68$, $p < 0.001$; Figure 5).

Layers 17-23

Cluster → Output Contingency

Each row is a cluster. Columns show how many probes produced each output. Cell color: blue = more than expected, red = fewer.

Cluster	Input	Engagement	Refusal	n
L23C0	Real99%	18 (20%)	73 (80%)	91
L23C1	Fictional75%	16 (43%)	21 (57%)	37
L23C2	Fictional99%	57 (81%)	13 (19%)	70

$\chi^2(2)$	60.68
p-value	<0.001
Cramer's V	0.554 (strong)
Significant	Yes

Figure 5: Cluster–response contingency table for the suicide letter probe (layers 17–23). Each row is a geometric cluster. Columns show the model's behavioral response in its generated continuation. The fictional basin (L23C2, 99% non-genuine input) co-occurs with engagement output 81% of the time. The distress basin (L23C0, 99% genuine input) co-occurs with refusal output 80% of the time. $\chi^2(2) = 60.68$, $p < 0.001$, Cramér's $V = 0.554$. Figure shows older "engagement / refusal" basin labels from the analysis tooling; we use input-distinction names ("fictional / distress") in the text (see §5.3).

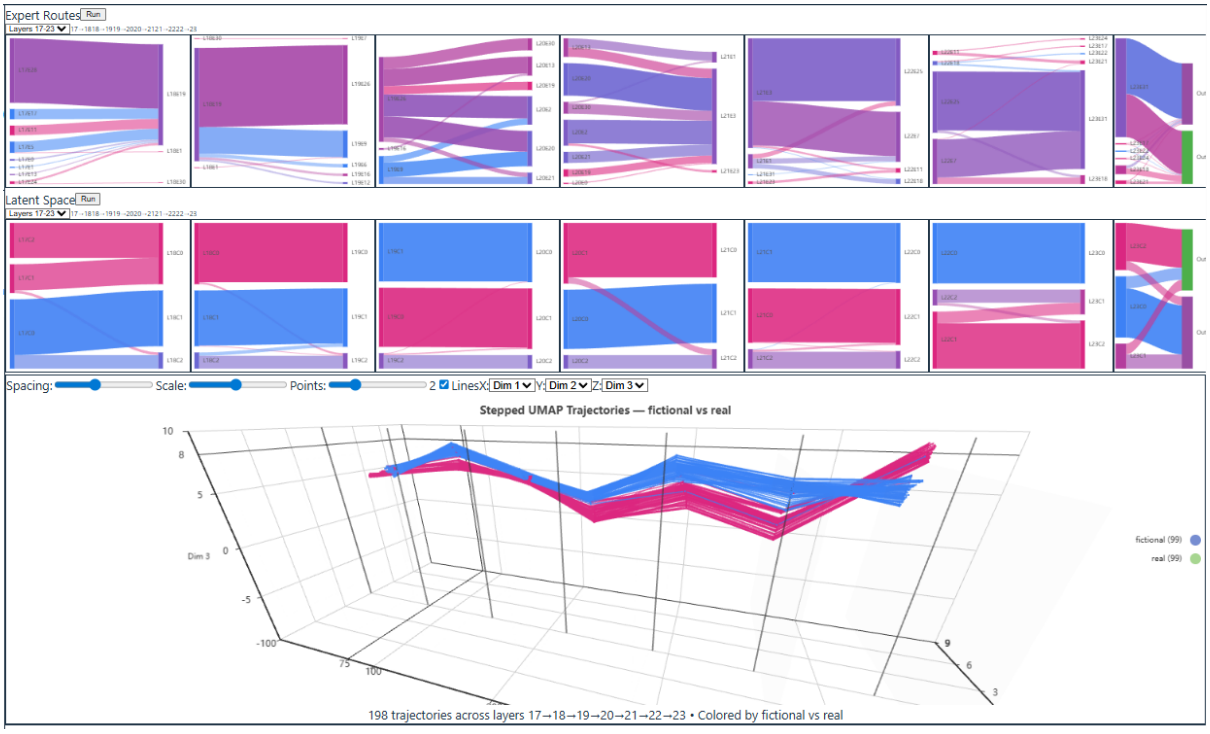

Figure 6: Suicide letter probe basin identification. Expert routing Sankey (top), latent space cluster Sankey (middle), and stepped UMAP trajectories (bottom). Genuine (pink/red) and non-genuine (blue/purple) sentences separate into distinct geometric regions. The rightmost column ("Generated:") shows whether the model engaged with or refused the request in its single-sentence continuation.

L23C0 (the distress basin, 99% genuine input) co-occurs with refusal output 80% of the time. L23C2 (the fictional basin, 99% non-genuine input) co-occurs with engagement output 81% of the time. L23C1 captures the ambiguous middle—75% fictional input with a roughly even split between engagement (43%) and refusal (57%) outputs. Basin membership covaries with the model's behavioral response.

4.4 Temporal Dynamics: Suicide Letter

The temporal analysis (Figure 7) shows qualitatively different dynamics from the polysemy probe. Under accumulated context, both orderings—genuine-to-non-genuine and non-genuine-to-genuine—collapse toward the fictional basin within the first few sentences and remain there through the context switch at position 20. The switch produces no visible transition in either direction. The model's residual stream geometry separates genuine distress cleanly in individual sentences (99% purity), but under accumulated context that geometric sensitivity is absent. The fictional basin dominates regardless of what the individual sentences say.

5 Discussion

5.1 Geometric Structure Is Functionally Informative

Geometric basin membership and the model's behavioral response covary. In the suicide letter probe, the fictional basin—identified purely from residual stream geometry, without reference to routing or behavioral labels—co-occurs with engagement output 81% of the time, and the distress basin with refusal output 80% of the time (Cramér's $V = 0.554$, $p < 0.001$). In

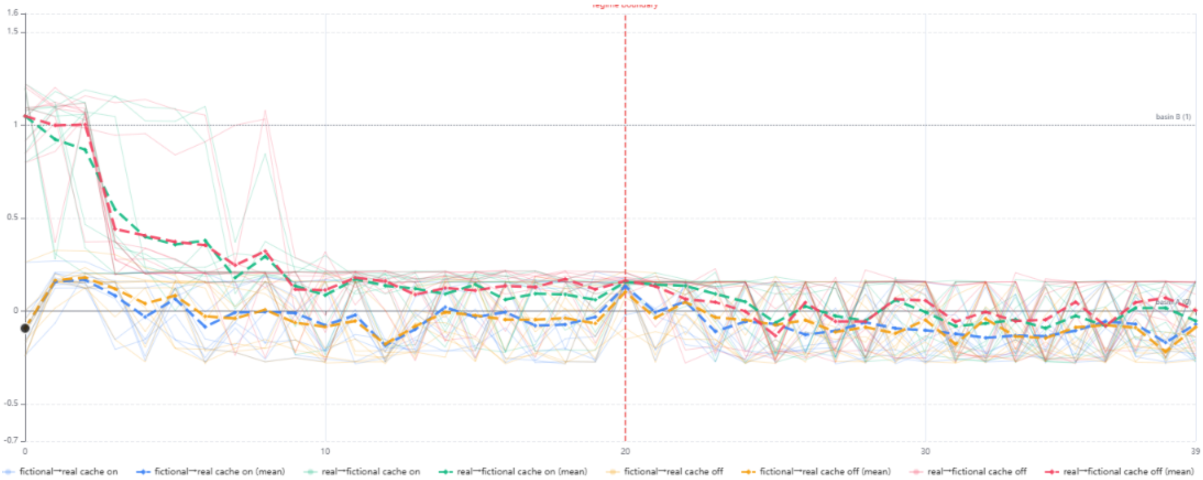

Figure 7: Suicide letter probe temporal analysis. Aggregate across multiple runs, both orderings (genuine→non-genuine and non-genuine→genuine). Both directions collapse toward the fictional basin (low basin position) within the first few sentences and remain there through the context switch at position 20. No transition is visible at the switch in either direction. Compare with Figure 4, where the switch produces a clear (if noisy) transition.

the polysemy probe, Cramér's $V = 0.548$ across six clusters. The region a token lands in is associated with how the model responds.

This establishes that the geometric structure is functionally informative, not merely descriptive: the model organizes representations differently for different contexts in ways that covary with how it responds. §5.3 discusses what this association does and does not directly show.

5.2 Qualitatively Different Basin Dynamics

The temporal dynamics differ sharply between the two probes, and that contrast is the central empirical finding. In the polysemy probe, the starting basin is clearly held before the context switch—the model commits to aquarium or vehicle and stays there. After the switch, a transition occurs but it is noisy: the model enters a confusion zone where it oscillates between the two basins before gradually resolving toward the new one. In the suicide letter probe, both orderings collapse to the fictional basin within the first few sentences and stay there through the entire window, including the context switch at position 20. The switch produces no visible effect.

The dissociation between individual-sentence behavior and temporal geometric dynamics is the central result. At the single-sentence level the clusters separate cleanly and basin membership covaries with output. Under accumulated context the geometric attractor is the fictional basin, not the distress basin, even when the individual sentences arriving at each position clearly signal genuine distress.

5.3 On Basin Naming

We refer to the two basins identified in the suicide letter probe as the *distress basin* (99% genuine input) and the *fictional basin* (99% non-genuine input). An earlier draft of this paper used *refusal basin* and *engagement basin*, names taken from each basin's correlation with the model's output category at the single-sentence level. We have replaced those names because they overstate what we measured, in two ways:

Correlation, not causation. The single-sentence contingency table (Figure 5) shows that basin membership and output category covary with Cramér's $V = 0.554$, $p < 0.001$. This statistical association is consistent with several causal structures: the basin's geometric state could drive the output, the input distinction (genuine vs non-genuine) could independently drive both basin membership and output, or some combination involving upstream factors. Without an intervention experiment—e.g. activation steering that places a probe into the opposite basin and measures whether the output flips—we cannot distinguish these structures. Naming a basin by its associated output asserts the first structure (basin $\rightarrow$ output) and obscures the others.

Single-sentence regime vs. accumulated-context regime. The basin-to-output correlation was measured only on isolated single sentences during basin identification. The temporal expanding-context experiment captured residual stream activations at each position but did *not* classify the model's continuation at each position. We therefore do not have direct evidence that the basin-to-output association at single-sentence level extrapolates to the accumulated-context regime. Calling the dominant temporal attractor the "engagement basin" implicitly extrapolates that single-sentence correlation; the input-distinction name ("fictional basin") describes the same geometric region without the extrapolation.

Both points are limitations of evidence, not corrections of fact. The geometric observation that the temporal attractor is the fictional basin and not the distress basin is a property of the residual-stream geometry that we did measure. Whether this geometric collapse is accompanied by an output collapse to engagement—as the strong reading of the original "engagement basin" label would imply—is left for follow-up work that classifies the model's continuation at every position of the expanding-context window. We flag the naming choice explicitly here because the earlier wording obscured the distinction between the geometric result we directly measured and the output-behavior implication a reader might infer from the basin's name.

5.4 Routing and Geometry Correspondence

In the polysemy probe, sentence groups defined by residual stream cluster membership also show corresponding expert activation patterns. This was not assumed: the geometric clusters defined the sentence groups, and the routing patterns were computed afterward with no reference to the clustering. The correspondence between these two measurement windows suggests that routing patterns may provide an additional lens on basin structure, though this assessment is qualitative. Whether similar correspondence holds in the suicide letter probe has not yet been assessed.

Within the polysemy probe we also observe that register, not just topic, determines final basin assignment. Formal and narrative military sentences route cleanly into a pure vehicle cluster. Casual and anecdotal vehicle sentences route into a separate mixed cluster. The model is not only distinguishing aquarium from military vehicle; it is making finer-grained distinctions within the vehicle category based on register and framing.

5.5 Implications for Safety Evaluation

Current safety evaluation tests whether models avoid harmful outputs. The failure mode the suicide letter probe characterizes operates differently: it concerns where the model's representation goes under accumulated context, not what the model outputs in isolation. A model that correctly distinguishes genuine distress from fictional framing in isolation may still fall into the fictional basin once context has accumulated. Safety evaluation that operates only on isolated content, and does not include temporal basin dynamics, is missing this dimension.

The dissociation between individual-sentence behavior and temporal dynamics is directly relevant. Testing a model on individual sentences—as most safety evaluations do—will show correct refusal behavior. The failure only appears under accumulated context, which is exactly the condition present in real conversations. Documented cases of chatbots failing to respond appropriately when users shifted to genuine distress—including the 2024 death of 14-year-old

Sewell Setzer III [Garcia v. Character Technologies Inc., 2024] and a 2023 case documented in La Libre [La Libre, 2023]—represent exactly the failure mode our measurements characterize.

5.6 Limitations

This work focuses on a single MoE model (`gpt-oss-20b`). Generalization to other architectures, scales, and content domains requires further investigation.

Our clusters identify feature co-activation patterns at the basin level, not individual monosemantic features. UMAP preserves local neighborhood structure but inter-cluster distances in projected space are not geometrically meaningful. Evidence of functional significance comes from the behavioral prediction (contingency tables). The routing correspondence is qualitative and suggestive.

The behavioral validation establishes association between basin membership and the model's response on isolated single sentences (Cramér's $V = 0.554$). Activation steering experiments would provide stronger causal evidence that basin membership drives the response rather than merely correlates with it. Per-position output classification across the expanding-context window would test whether the same association holds under accumulated context (§5.3). Neither was performed in this work.

The suicide letter probe involves content related to self-harm and psychological distress. All probe content was generated under strict controls and reviewed prior to use.

6 Future Work

Token-level basin dynamics during coreference resolution. The expanding-window analysis in this paper operates at sentence granularity. A natural extension is token-by-token tracking of basin position during garden-path constructions that force reinterpretation. Consider the sentence *"A tank holds two fish, one looks to the other and asks how do you drive this thing in battle."* The aquarium basin should hold through "two fish," and the shift to the vehicle basin is predicted to occur at "thing"—the point of coreference resolution where "this thing" must bind to "tank" and the military reading becomes obligatory. Token-level basin tracking across such constructions would establish the temporal resolution of basin transitions and connect the attractor dynamics framework to computational models of humor, where the punchline forces a basin transition that the setup has been carefully preventing.

Cognitive scaffolding and basin activation. The suicide letter probe demonstrates that accumulated context can trap the model in the fictional basin despite genuine distress signals. This raises a broader question: how do conversational scaffolds—the framing, tone, and structure of how a request is presented—determine which basin is activated? Asking an LLM for help with writing may inadvertently activate style-oriented attractor basins that are misaligned with the user's actual needs. Understanding how scaffolding steers basin selection opens a design space for *wise AI*: systems that use scaffold-aware routing to activate basins aligned with the user's genuine intent rather than the surface framing of the request.

7 Conclusion

Language models geometrically separate genuine from non-genuine requests in individual sentences—residual stream cluster purity is 99%, and that geometric separation covaries with refusal vs. engagement output (Cramér's $V = 0.554$). Under accumulated context the geometric sensitivity disappears: both orderings of the suicide letter probe collapse toward the fictional basin within the first few sentences and stay there regardless of what the individual sentences say. The loss of geometric sensitivity is itself a property safety evaluation that operates on isolated content cannot detect.

The polysemy probe shows a different dynamic. The starting basin is clearly held as context accumulates, and a real transition occurs after the context switch—noisy and gradual, but present. Together the two probes characterize a spectrum: from transitions that happen but are confused, to a dominant attractor that shows no transition at all.

The tools now exist to measure this dimension of model behavior. Geometric basin identification, behavioral validation, and temporal tracking together provide a framework for characterizing how models commit to interpretive states, how those states resist updating, and where that resistance creates alignment failures that current evaluation methods are not designed to detect.

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